

From Predictive Uncertainty to Decision-Effective Dynamics Ambiguity Sets

Robust Control of Function-Valued Systems under Deployment Shift

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Abstract

Learned dynamics can be calibrated in a predictive norm yet still induce poor control when errors align with sensitive cost-to-go directions. We ask how predictive uncertainty can become a decision-effective dynamics ambiguity set. A function-valued world model outputs a mean and pointwise scale; a small deployment audit conformalizes the resulting anisotropic ellipsoid. Projected gradient ascent over this set gives a nonlinear adversarial controller, while its closed-form support in the finite-horizon adjoint direction gives a fast first-order approximation. We establish marginal one-step coverage, derive the exact support and linearization remainder, and state rollout and discounted value-error bounds under Lipschitz assumptions. Controlled viscous Burgers dynamics test viscosity, boundary, and latent actuator-gain shifts. In a preliminary three-seed paired gain sweep, adversarial MPC reduced empirical 90th-percentile cost in every seed by 1.57%–3.68%; the mean relative change in average cost across seeds was +0.05%.

1 Introduction

Fast learned surrogates make repeated planning for distributed-parameter systems increasingly plausible. Neural operators, for example, learn function-to-function maps and can replace expensive numerical time steppers in model predictive control (MPC) [6]. Their deployment creates a separate difficulty: a controller optimizes against a learned model and can therefore exploit precisely those errors that appear harmless under an average prediction metric.

Uncertainty quantification does not resolve this problem by itself. Calibration is a statement about an event and a score; it does not state that the covered directions matter for a decision. This distinction is particularly sharp for a function-valued state. A large error far from a controlled region can have little effect on the objective, whereas a smaller error aligned with a cost-to-go gradient can reverse the preferred action. Related work has shown that calibrated uncertainty improves model-based planning [7], that model losses can be made value-aware [4], and that uncertainty estimates for PDE surrogates can degrade under moderate distribution shift [8]. These facts motivate a narrower question:

How can predictive uncertainty be converted into a dynamics ambiguity set that is effective for control decisions?

Conformal regions have already been combined with controller robustification [2], robust optimization [5], ensemble-based OOD adaptation [3], and OOD system-level synthesis [9]. We therefore do not claim that conformal robust control is new. Our target is the geometry connecting a calibrated function-space dynamics set to the downstream finite-horizon objective.

The proposed construction begins with a heteroscedastic predictive ellipsoid, calibrated on a small deployment audit. Robust MPC then solves an approximate inner maximization over a rectangular product of these one-step sets. Automatic differentiation also computes the adjoint of the remaining nominal cost, giving a closed-form first-order alternative. This separates three effects that are often conflated: using deployment data, obtaining predictive coverage, and improving the resulting decision.

Our intended contributions are:

1. an audit-calibrated, anisotropic dynamics ambiguity set for function-valued transitions;
2. a nonlinear adversarial-rollout controller and an exact adjoint support formula for its first-order approximation;

3. a transparent hierarchy of finite-sample, deterministic, and first-order theoretical statements; and
4. an evaluation protocol that requires closed-loop benefit beyond coverage, using matched in-distribution, audit-L2, and adjoint-support baselines.

The initial implementation uses a residual Fourier neural operator (FNO) as a world model. The paper is not an FNO architecture contribution.

2 Problem formulation

Let $x_t \in \mathbb{R}^n$ be a discretized function and $a_t \in \mathcal{A}$ a control. The unknown deployment dynamics are

$$x_{t+1} = G_\star(x_t, a_t; \xi),$$

where ξ includes physical parameters, boundary conditions, and latent actuator properties. A learned model is trained on a source distribution, then used to minimize the finite-horizon cost

$$J_H(x_0, a_{0:H-1}) = \sum_{t=1}^H \ell(x_t) + \sum_{t=0}^{H-1} r(a_t).$$

At deployment, we assume access to a small independent audit sample $\mathcal{D}_{\text{aud}} = \{(x_i, a_i, y_i)\}_{i=1}^m$, drawn from the deployment regime but not used to update the mean dynamics model.

We call an ambiguity set *decision-effective* if, at a matched audit budget and nominal model, incorporating it into the controller improves a predeclared downstream metric: paired mean closed-loop cost, upper-tail cost, or constraint violation. Predictive coverage is necessary evidence, but is not the definition.

3 From predictive uncertainty to a dynamics ambiguity set

3.1 Function-valued mean and scale

The world model outputs

$$\hat{G}_\theta(x, a) = \mu_\theta(x, a), \quad \sigma_\theta(x, a) \in \mathbb{R}_{\geq 0}^n.$$

Our prototype uses residual operator blocks and predicts $\mu_\theta(x, a) = x + \delta_\theta(x, a)$. The scale head is trained by a pointwise quantile loss on detached absolute residuals, preventing the mean model from increasing its error merely to simplify uncertainty fitting.

3.2 Deployment-audit conformalization

Write $\|v\|_{2,n}^2 = n^{-1} \sum_{j=1}^n v_j^2$. Define the audit score

$$S_i = \left\| \frac{y_i - \mu_\theta(x_i, a_i)}{\sigma_\theta(x_i, a_i)} \right\|_{2,n}.$$

For target miscoverage α , let q be the empirical order statistic with rank $\lceil (m+1)(1-\alpha) \rceil$, clipped at m . The ambiguity set is

$$\mathcal{U}_\theta(x, a) = \left\{ \mu_\theta(x, a) + \sigma_\theta(x, a) \odot z : \|z\|_{2,n} \leq q \right\}. \quad (1)$$

Proposition 1 (Marginal one-step coverage). *Conditional on the trained model, if the audit examples and a new deployment example are exchangeable, then*

$$\Pr\{G_\star(x, a) \in \mathcal{U}_\theta(x, a)\} \geq 1 - \alpha.$$

Proof. The m audit scores and the new score are exchangeable. The rank of the new score is therefore uniform up to ties; the conformal order statistic gives the stated lower bound. $\square$

4 Decision-effective robust MPC

4.1 Nonlinear adversarial rollout

For a candidate action sequence, the rectangular robust planning problem is

$$\min_{a_{0:H-1}} \max_{\|z_t\|_{2,n} \leq q} \sum_{t=1}^H \ell(x_t) + \sum_{t=0}^{H-1} r(a_t), \quad x_{t+1} = \mu_\theta(x_t, a_t) + \sigma_\theta(x_t, a_t) \odot z_t. \quad (2)$$

The prototype uses projected gradient ascent for $z_{0:H-1}$ and the cross-entropy method for the outer action sequence. This computes an approximate saddle point; finite PGD iterations are not an optimality certificate. The rectangular product is a robust planning choice. Marginal one-step conformal coverage does not imply simultaneous coverage of every set along a trajectory.

4.2 Fast adjoint approximation

Use $\langle v, w \rangle_n = n^{-1} \sum_j v_j w_j$. For a candidate nominal rollout, let

$$\lambda_{t+1} = \nabla_{x_{t+1}} \hat{J}_H$$

be the full finite-horizon adjoint, including the dependence of later predicted states on x_{t+1} .

Proposition 2 (Exact ellipsoid support). *For every $\lambda \in \mathbb{R}^n$,*

$$\sup_{\Delta: \|\Delta/\sigma\|_{2,n} \leq q} \langle \lambda, \Delta \rangle_n = q \|\lambda \odot \sigma\|_{2,n}. \quad (3)$$

Proof. Set $\Delta = \sigma \odot z$. Cauchy–Schwarz gives $\langle \lambda \odot \sigma, z \rangle_n \leq \|\lambda \odot \sigma\|_{2,n} \|z\|_{2,n}$. Equality is attained by choosing z parallel to $\lambda \odot \sigma$. $\square$

Linearizing the inner problem in Equation (2) gives

$$\hat{J}_H(a_{0:H-1}) + \sum_{t=0}^{H-1} q \|\lambda_{t+1} \odot \sigma_\theta(\hat{x}_t, a_t)\|_{2,n}. \quad (4)$$

This adjoint-support controller is retained as a computationally cheap approximation and ablation. The nonlinear adversarial rollout is the primary robust controller.

4.3 Linearization remainder

Proposition 3 (Conditional curvature bound). *Suppose the Hessian of the remaining horizon cost with respect to x_{t+1} has operator norm at most M_H throughout $\mathcal{U}_\theta$. Then*

$$\sup_{\Delta \in \mathcal{U}_\theta} J_H(\hat{x} + \Delta) \leq J_H(\hat{x}) + q \|\lambda \odot \sigma\|_{2,n} + \frac{M_H}{2} q^2 \|\sigma\|_\infty^2.$$

The adjoint controller uses the first-order terms in Equation (4); the adversarial controller instead optimizes the nonlinear rollout numerically. Neither is trajectory-certified: the former omits a certified curvature buffer, while the latter uses a finite-step nonconvex inner solver.

5 Rollout and value-error bounds

Assumption 1. *On a forward-invariant set, $G_\star$ is L_G -Lipschitz in state and $\|G_\star(x, a) - \hat{G}_\theta(x, a)\|_{2,n} \leq \epsilon$ uniformly.*

For a common open-loop action sequence and $e_t = \|x_t - \hat{x}_t\|_{2,n}$,

$$e_{t+1} \leq L_G e_t + \epsilon, \quad e_h \leq \epsilon \sum_{j=0}^{h-1} L_G^j.$$

If the stage reward is L_r -Lipschitz and $\gamma L_G < 1$, then

$$|V_{G^*} - V_{\hat{G}}| \leq \frac{\gamma L_r \epsilon}{(1 - \gamma)(1 - \gamma L_G)}.$$

If every policy in the comparison class induces true and learned closed-loop maps sharing the same Lipschitz bound, a two-model optimality decomposition gives

$$V_{G^*}^{\pi^*} - V_{\hat{G}}^{\hat{\pi}} \leq \frac{2\gamma L_r \epsilon}{(1 - \gamma)(1 - \gamma L_G)}.$$

For $L_G \leq 1$, this has the desired $O(\epsilon/(1 - \gamma)^2)$ dependence. These are deterministic uniform-error bounds; marginal conformal coverage alone does not imply their assumptions. Related Lipschitz model-based bounds are developed in [1].

6 Experimental design

System. We use the controlled viscous Burgers equation

$$u_t + uu_x = \nu u_{xx} + g_{\text{act}} a_t b(x),$$

with nonhomogeneous Dirichlet boundaries. Training and deployment differ in viscosity, boundary values, initial-condition amplitude, and the latent actuator gain g_{act} . The controller observes the state and boundary condition but not g_{act} .

Baselines. All methods share one trained residual FNO: (i) uncontrolled, (ii) nominal MPC, (iii) an L2 tube calibrated on source data, (iv) an L2 tube calibrated on the same deployment audit, (v) the audit-calibrated ellipsoid with adjoint support, and (vi) the same ellipsoid with adversarial nonlinear rollouts. The comparison between (iv)–(vi) isolates ambiguity geometry and inner-solver effects from access to deployment data.

Metrics. We report one-step set coverage, radius–error association, mean and median closed-loop cost, 90th-percentile and worst-case cost, paired seed-level differences, and action magnitude. A positive result requires that the proposed method improve at least a predeclared tail-risk metric without a material increase in mean cost. Model RMSE alone cannot establish the claim.

6.1 Preliminary diagnostic: why coverage is insufficient

Before introducing the conformal ellipsoid in Equation (1), we tested scalar L2 and fixed spatially weighted radii on three shifted rollouts. Deployment audit calibration increased combined-shift one-step coverage from 0.611 to 0.822–0.856, but no robust controller improved on the uncontrolled cost. This small smoke test is not a final comparison. It serves as a falsifying diagnostic: restoring predictive coverage did not by itself produce a useful decision. Quantitative claims remain preliminary until the paired study includes substantially more initial states.

6.2 Preliminary paired gain sweep

We next fixed the initial field, viscosity, and boundary conditions within each seed and evaluated all controllers at latent actuator gains $\{-0.5, 0, 0.5, 1, 1.5\}$. This removes the confounding between shift severity and initial-condition difficulty. Table 1 reports three initial seeds. The audit contains 90 independent deployment transitions.

The adversarial controller improves the empirical upper-tail metric in all three seeds with negligible average change in mean cost. On the same calculation, the audit-calibrated L2 tube improves p_{90} by 1.74%. This supports the mechanism but not a population claim: each seed contains only five paired gains, and the nonlinear inner maximization is approximate.

Table 1: Preliminary paired-factorial evidence. Changes are relative to nominal MPC within each seed. Empirical p_{90} uses only five gain levels.

Seed	ID coverage	Audit coverage	Mean change	p_{90} change
0	0.283	0.922	+0.41%	−3.68%
1	0.206	0.950	−0.41%	−1.59%
2	0.222	0.878	+0.15%	−1.57%
Mean relative change	−	−	+0.05%	−2.28%

7 Limitations and planned stress tests

Split conformal validity requires exchangeability between the audit and target deployment examples. The guarantee is marginal and one-step; it is not a simultaneous trajectory guarantee. The robust objective is first order unless the curvature buffer is certified. The pointwise scale can be misspecified; a deep FNO ensemble is therefore a required uncertainty baseline [8]. Finally, one-dimensional Burgers control is a mechanism test, not evidence of broad PDE-control safety. Planned stress tests include audit-size sweeps, shift-severity sweeps, adversarial actuator degradation, paired random seeds, and a second advection–diffusion system.

8 Conclusion

Predictive uncertainty becomes useful for control only after its geometry is connected to the downstream objective and verified in closed loop. The proposed route is deliberately simple: conformalize one anisotropic dynamics ellipsoid, then optimize control against adverse transitions in that same set. The adjoint support provides a fast, interpretable approximation; projected adversarial rollout retains the nonlinear cost. Whether either conversion improves decisions is an empirical question, and the evaluation is designed to reject the method when it improves coverage but not control.

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